

Supplemental Material:

Can a Neural Encoding Model Replicate an fMRI Visualization Study?

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A Time Step Comparison

Table 1: Time step for frame extraction versus the number of matches with the prior study, **Color**.

Time Step	# Matches
$t = 0$	7
$t = 1$	7
$t = 2$	8

Table 2: Time step for frame extraction versus the number of matches with the prior study, **Grayscale**.

Time Step	# Matches
$t = 0$	4
$t = 1$	4
$t = 2$	3

B De-Centered Results

Table 3: Comparison of the model’s predicted contrast (primary) and prediction demeaned compared to the prior study, for *Experiment 1* (color). Positive values indicate higher predicted fMRI response for **Surface** . A  symbol indicates alignment with the corresponding result of the prior study, while  indicates disagreement. L stands for *left* and R stands for *right*. Note that, based on the demo provided in the TRIBE v2 repository, the value range for model responses is roughly $[-1, 1]$.

Cluster (Walden)	Prior Result	Our Result	Demeaned
R Temporal Occipital Fusiform	Surface 	+0.041 	+0.047 
L Temporal Occipital Fusiform	Surface 	+0.019 	+0.025 
L Inferior Lateral Occipital	Surface 	−0.030 	−0.023 
L Superior Lateral Occipital	Surface 	+0.056 	+0.063 
L Superior Parietal Lobule	Bubble 	+0.038 	+0.045 
L Middle Frontal Gyrus	Bubble 	−0.011 	−0.005 
L Occipital Pole	Bubble 	−0.087 	−0.080 
R Middle Precentral Gyrus	Bubble 	−0.021 	−0.015 
R Anterior Cingulate	Bubble 	−0.017 	−0.011 

Table 4: Grayscale (Experiment 2). Predicted contrast at prior study’s reported cortical clusters. Positive values indicate **Surface** -dominance. A  symbol indicates matching with the results of the prior study, while a  symbol indicates disagreement. L stands for *left* and R stands for *right*. Note that, based on the demo provided in the TRIBE v2 repository, the value range for model responses is roughly $[-1, 1]$.

Cluster (Walden)	Prior Result	Our Result	Demeaned
L Superior Lateral Occipital	Surface 	+0.142 	+0.112 
L Middle Frontal Gyrus	Surface 	+0.012 	−0.019 
R Superior Frontal Gyrus	Surface 	−0.011 	−0.042 
R Fusiform Cortex	Surface 	+0.095 	+0.064 
R Cuneus	Bubble 	−0.049 	−0.079 

C Example Results Visualization

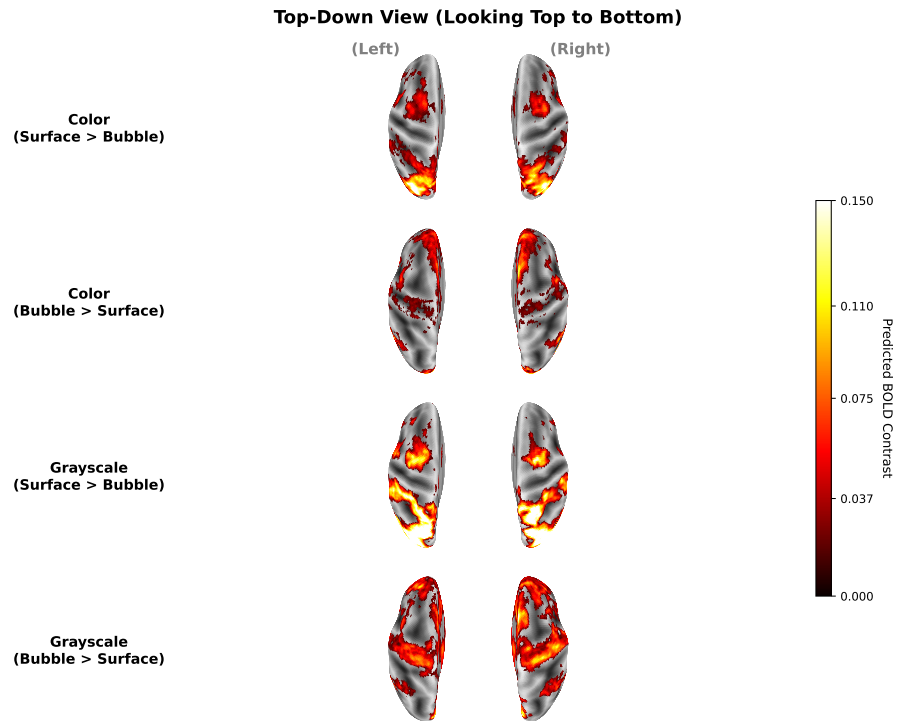


Figure 1: Predicted BOLD contrast on the fsaverage5 Surface $\blacklozenge$ (dorsal view). Positive contrast (Surface $\blacklozenge$ – Bubble $\blackstar$) and negative contrast (bubble – Surface $\blacklozenge$) are plotted separately for each experiment, mirroring the convention of Walden et al.’s Figures 9 and 10. A single color scale (0 to 0.15) is shared across all four panels. The prominent clusters along the posterior-lateral boundaries correspond to occipital visual areas.

D Strongest Predicted Regions

Table 5: Strongest predicted regions in each experiment (top five per direction, Destrieux parcels with ≥ 50 vertices).

Region	Contrast
Grayscale: Surface  > Bubble 	
R Middle occipital / lunate sulcus	0.251
L Superior / transverse occipital sulcus	0.222
L Middle occipital / lunate sulcus	0.212
L Middle occipital gyrus	0.204
R Superior / transverse occipital sulcus	0.195
Grayscale: Bubble  > Surface 	
R Cuneus	0.049
R Central sulcus	0.035
R Inferior frontal (triangular)	0.032
L Anterior insular sulcus	0.030
R Anterior insular sulcus	0.025
Color: Surface  > Bubble 	
R Lingual / medial occipito-temporal sulcus	0.140
L Lingual / medial occipito-temporal sulcus	0.113
L Superior / transverse occipital sulcus	0.105
R Superior / transverse occipital sulcus	0.085
R Parieto-occipital sulcus	0.072
Color: Bubble  > Surface 	
L Occipital pole	0.087
R Occipital pole	0.070
R Superior temporal gyrus (lateral)	0.064
R Superior temporal sulcus	0.062
R Inferior frontal (triangular)	0.061